

RESEARCH ARTICLE

Multi-Cluster Graph (MCG): A Novel Clustering-Based Multi-Relation Graph Neural Networks for Stock Price Forecasting

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ABSTRACT Stock price forecasting is one of the indispensable challenges, as financial markets are increasingly becoming volatile due to the complex, noisy, dynamic, and chaotic nature of stock data. Accurate stock price forecasting models play a key role in protecting risks associated with markets and making sound investing decisions. Mostly, existing studies suggested different Machine Learning (ML) and Deep Learning (DL) based methodologies, but they often focus on individual stock data to generate predictions, ignoring the interconnectivity and interconnections that exist between stocks. Recent advances in graph neural networks highlighted the need to model stock-to-stock interactions to improve the reliability of forecasting. Regardless of the potential, modeling these interactions among stocks poses substantial challenges. Existing methods encompass basic heuristics like sector affiliations, shareholdings, or co-occurrence in news headlines, etc., to model relationships among stocks which can disregard the intricate and varied nature of stock interconnection. To address such issues, this study proposed a clustering-based approach to exploit relationships among stocks listed on particular stock exchanges. More precisely, different graph models are created using three proposed strategies including RS (Returns-Sharpe Ratio), VR (Volatility-Return), and YR (Yearly variance and returns) clustering. These clusters are employed to create multi-relation graphs which are then propagated to an LSTM-based graph network to forecast future prices at the node level. The proposed approach can anticipate future stock prices for all stocks featured in an NSADAQ and S&P500 index while utilizing graph nodes from a single-trained standalone model. Moreover, experimental results on NSADAQ-100 and S&P500 stocks, show that the proposed clustering-based stock graph models methodology shows good results in terms of lowest MSE and MAE values of 0.01 and 0.08 respectively.

INDEX TERMS Stock price forecasting, deep learning, graph neural networks, LSTM, multi-relation graphs, stock clustering strategies.

I. INTRODUCTION

The stock market is a system of financial instruments that facilitates transactions among corporations and investors, with a worldwide value of over \$83.5 trillion as of 2020 [1]. According to EMH (Efficient Market Hypothesis), stock prices reveal important information, however, their volatile nature makes the challenge of forecasting price movement difficult [2]. These swings in stock prices are due to sev-

eral reasons including factors affecting the economy such as macroeconomic trends, cycles, structure, industry growth, and financial health of the companies listed [3], [4]. Besides that, some other factors include the psychological conditions of investors and micro factors [5]. Researchers as well as investors are looking for opportunities to profit from stocks. As a result, forecasting techniques for stocks are frequently used [6]. Hence, for many years, investors have been seeking reliable information about future stock prices and movements to make informed investment decisions. Since the early 1900s, the financial sector has invested millions of dollars

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in developing and perfecting price prediction models [7]. To provide time series prediction results that are more precise, the prediction task seeks to build a relationship model between past financial data as well as future patterns in stock prices. Traditional techniques of stock market forecasting involve technical and fundamental analysis. Technical analysis analyzes stock price history as well as indicators to anticipate future trends, whereas fundamental analysis looks at the fundamental value as well as is often utilized by human specialists. For instance, a group of studies attempts to forecast future prices of stocks with the help of regression methods [8], while others attempt to solve this problem as a classification problem in which stock movements are predicted [9]. The most popular algorithms in all of these are ML (Machine Learning) models, DL (Deep Learning Algorithms) [10] such as LSTM (Long-Short Term Memory) [11] and advanced GRU-based (Gated Recurrent Units) methods [12], and hybrid methods [13]. These deep learning methods have a great advantage over classical forecasting methods such as ETS (Exponential Smoothing) and ARIMA (Autoregressive Integrated Moving Average). To address the limitations of different techniques and improved forecasting results, some research studies have involved optimization techniques [14], feature selection methods [15], and attention mechanisms [16]. The utilization of text mining approaches with stock price forecasting is also investigated and exploited such as integration of news sentiment scores along with multi-layer neural networks for regression tasks [17]. This inclusion provides for a more comprehensive knowledge of market behavior since sentiment scores reveal information on investor attitude, market sentiment, and the market's general sentiment. Some studies have integrated the events sentiments such as COVID-19 [18], and Brexit [19], and Election events [20] that have a major impact on stock prices. This integration enables a more thorough examination of the elements influencing stock price movements in reaction to big political or economic events.

However, in most of the research studies, prediction models rely on historical data of just one stock to forecast future values, without considering inter-stock interactions. This is due to the reason that the main premise and theory of these methodologies is that stocks are independent of one another. The methodologies neglect the cross-impact of stocks as time passes which can have a dynamic influence on stock prices [21]. Shared industry knowledge, supply chains, payment networks, company partnerships, and shareholder ownership all contribute to cross-stock interactions [22], [23], [24].

Individual stock prices have a strong correlation with other stocks due to complicated and varied relationships, in addition to their own information. For example, in some studies, it is observed that the stock prices of one stock can lead or lag the stock prices of other stocks [22]. Hence, it is quite important to take into account the stock relationships while predicting their future prices more optimally. It is an important argument that representation in the form of a time series for stock price predictions cannot capture the entire

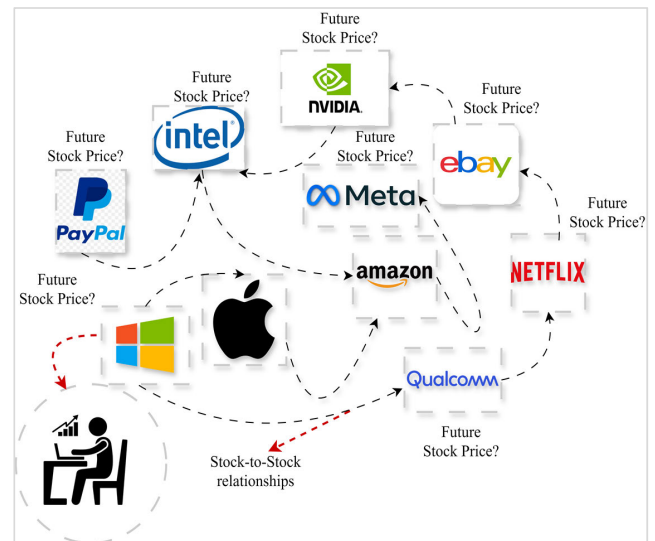


FIGURE 1. Graph models indicating stock to stock relationships.

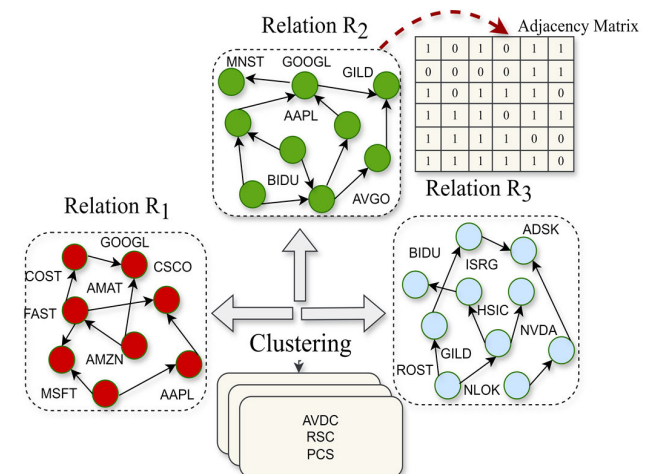


FIGURE 2. Multi-relation graph models based on proposed clustering approaches.

information because of the missed relation between stocks to stock. Moreover, these conventional methods of stock market predictions perceive the time series as independent and evenly spread among themselves, which is inconsistent with the reality of the financial market [25]. For instance, two stocks that belong to identical sectors have higher values of correlation than those that belong to different sectors [24].

Hence, in the recent past, another emerging trend in which interconnections of financial instruments are exploited. For instance, markets as well as stocks are inherently related entities interacting with one another. This idea is exploited in the latest research studies with the use of graph neural networks [21]. Initially, stock price synchrony in 1988 was studied [26], followed by several studies and empirical research [27], [28] that have demonstrated these cross-effects, in fact, cross-effects across distinct companies that have specific

correlations [29]. Researchers have recently improved stock price prediction by using graph structures to draw interactions connected to stocks, with the advancement of these theories and graph representation learning methods. To be more specific, a financial ecosystem is a kind of framework that has many elements and complicated relationships that are usually updated. To formulate and depict financial data in the field of finance, different types of graphs are designed such as graphs representing transaction network [30], user-item feedback graph [31], as well as stock-relations graphs() [32]. In order to convert financial problems into node classification problems, the usage of Graph Neural Networks (GNN) is commonly employed because they show good results in graph modeling techniques [33]. These GNN methods can be utilized for stock prediction problems in which each node symbolizes a stock, and the edge denotes the relationships among stocks. The simple graph model showing relationships among stocks is depicted in Figure 1. Each node in the graph represents a specific stock, and the goal was to estimate the future stock price of each stock by analyzing the relationship between stocks.

The complicated nature of financial frameworks involves the requirement of several data sources as well as complex graph models that lead to major challenges in graph building, and feature processing, alongside GNN modeling [21]. Furthermore, it is challenging to create a graph that accurately depicts the multifaceted nature of financial connections. Since the graph associated with finance is frequently diverse or time-varying, it presents challenges for GNN models that are presently in usage [21]. Existing studies have proposed graph-based models for stock market prediction, however, most of the studies have done stock movement prediction, which is quite easy, instead of stock price prediction which is more challenging. Furthermore, existing graph-based models build stock-to-stock relationships using some pre-defined criteria, e.g. sector-based relationships, the occurrence of both stocks in news headlines, and or based on some other criteria [24], [34]. These relationships could overlook the complete range of interactions between stocks, resulting in sparse adjacency matrices because two stocks may have similar volatilities regardless of their sector. Hence, this research presents a new clustering-based multi-relation graph neural network for stock-price prediction. Clustering can reveal hidden patterns in the movement of stocks that are not readily obvious from sector assessment. For example, stocks in distinct sectors may react similarly to macroeconomic changes, that clustering can emphasize. More precisely, three clustering strategies are proposed named RS (Returns with Sharpe Ratio), VR (Volatility-Return Clustering), and YR (Yearly variance and returns) clustering. This clustering leads to stock clusters that are further employed to build multi-relation graphs among stocks. Such information is processed at the node level using graph convolution layers followed by time-series forecasting using LSTM-Graph neural networks. The pin-point contributions of this work are as follows:

- A novel multi-relation graph neural network entitled Multi-Cluster Graph (MCG) has been proposed for predicting market prices.
- Three clustering approaches named RS (Returns with Sharpe Ratio), VR (Volatility-Return), and YR (Yearly Variance and returns) clustering are proposed to generate multi-relation graph models of stocks
- The proposed standalone model forecasts future stock prices at the graph node level
- Experimentation with 100 stocks on the NASDAQ 100 index and 500 stocks on the S&P500 index demonstrates that the proposed MCG model shows superior performance in terms of MSE and MAE.

The remainder of this article has been divided into different sections: Section II provides the related work, Section III discusses the proposed work, and Section IV shows the experimental results with a discussion, followed by Sections V and VI, which provide a comparative analysis, limitations, future directions, and conclusions. The pictorial representation of stocks in a graph model using the proposed clustering approach is depicted in Figure 2.

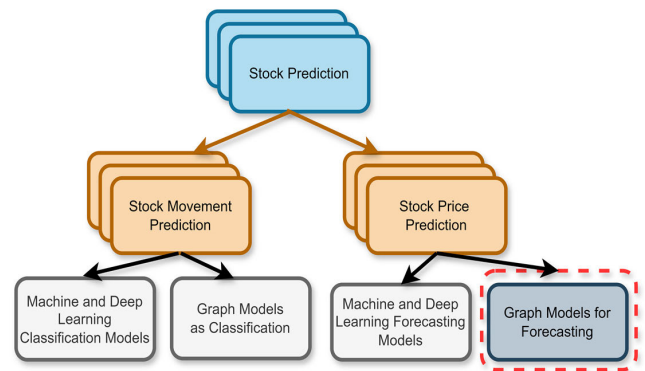


FIGURE 3. Pictorial overview of stock prediction models in literature.

II. RELATED WORK

This section provides a detailed literature review on stock price forecasting strategies classified as traditional methods, deep learning methods, and most lately graph models with a shed light on problems and highlighting research gaps with respect to the proposed work. The pictorial overview of existing methods is depicted in Figure 3.

A. TRADITIONAL MACHINE LEARNING METHODS

The traditional machine learning technique for stock price prediction is one of the oldest and most widely used approaches. More precisely, stock price prediction is quite a challenging task due to different environmental factors such as business financial performance, business data, public news, and societal sentiment. To perform stock prediction, different traditional machine learning methods have been proposed in the past. For instance, many studies rely heavily on historical market data, such as stock prices and trade activities

to perform prediction of stock prices. Popular algorithms include linear regression (LR) and support vector regression (SVR) models [35], [36], [37]. An improved version of the stock price prediction model is also proposed in [38]. More precisely, they have proposed an extended primal objective function to optimize the value of hyperparameters in SVR. Likewise, in [39], the bat-optimization algorithm is employed with SVR to optimize its three major free parameters. The experiments reveal that the BA-SVR model performs better than the polynomial kernel SVR method as well as the sigmoid kernel SVR. According to the features of the stock market that have both linear and nonlinear elements, and to integrate the strength of linearity and non-linearity, the ARIMA model is combined with SVR to carry out stock market forecasting [40]. When compared to solo ARIMA and SVR models, the combined model beats the individual models. Moreover, some studies approach the problem of stock price prediction as a classification problem in which they have employed utilized classification models. These models include random forest (RF), KNN, Naïve Bayes, and Support Vector Machines [41], [42]. To predict rising and downward trends of stock prices, a combination of SVM and fuzzy set theory is also exploited [43]. The major objective of such algorithms is to forecast future trends of stock prices as an upward trend or a downward trend. Other than such technical analysis-based methods, fundamental analysis of stock data is also carried out for stock market prediction. For this, research studies take into account a variety of elements, including yearly fiscal summaries and reports, balance sheets, income statements, etc [44]. Although these traditional methods show good results, however, they face difficulties in modeling the sequential nature of time series stock data, as well as heavily dependent on feature engineering methods [45]. Moreover, they also fail to model the interaction between stocks e.g. corporation relationship [24].

B. DEEP LEARNING METHODS

Other than machine learning methods, some deep learning methods have also been proposed for stock market forecasting similar to its tremendous use in other applications [46], [47], [48]. One of the most prominent and widely used methods is Recurrent Neural Networks (RNNs) and its advanced variants such as LSTM and GRU models [49]. In these deep learning-based methodologies, some advanced models are also proposed such as attention mechanisms [16]. For instance, historical data of different stocks are employed followed by using Bi-directional LSTM models (Bi-LSTM) to forecast the trends of stock prices [50]. The researchers provided a framework for quicker depth as well as improved learning compared to the unidirectional method. Some studies employed event sentiment such as COVID sentiment for stock market prediction using deep learning [18]. Since news coverage of key events can provide valuable insights into market drivers. Other than COVID, another event sentiment-based method is proposed in [19]. They have

studied the impact on stock prices of European Union (EU) nations per sentiments regarding the Brexit event by classifying nations as least contributing and major contributors. Moreover, instead of utilizing only the time series data of stock prices, some studies employ different data representations. For instance, in [51], two deep learning models with different data representations are employed. More precisely, the first model called LSTM works on time series data and extracts temporal features while the CNN model works on stock-chart images and extracts image features followed by predicting stock prices. Their LSTM-CNN model beats individual models at forecasting stock prices. Furthermore, their study shows that a candlestick chart is the most suitable stock chart visual to employ for predicting stock prices. Following on, in some research studies, hybrid models are also employed for stock price forecasting. For instance, in [52], a hybrid deep learning model reliant on Bidirectional Cuda Deep Neural Network LSTM (BiCuDNNLSTM) and a one-1D Convolutional Neural Network (CNN) is proposed for the prediction of stock prices. The reported findings suggest that their proposed hybrid model is effective for accurately predicting stock prices and reliable for assisting investors in making knowledgeable decisions. Although these methods show good results, by analyzing market signals for each stock individually. However, such approaches overlook the intrinsic relationships between stocks, resulting in inferior performance.

C. GRAPH MODELS

In the recent past, different approaches have been exploited by researchers to extend standard neural networks on graph-structured data. Because of the exceptional performance of CNN, the concept of convolution has been redesigned to deal with graph data, referred to as graph convolutions. Reliant on these convolutions, the underlying neural network called graph neural network has received a lot more attention from researchers because of its exceptional performance on node classification tasks [53]. More explicitly, graph models are provided with graph structure along with features of data as input on its nodes. Then, it can draw the complicated interactions among the nodes of the graph by summing up information from every node's neighbor followed by operating linear transformations. Such characteristics make these networks to be employed in extensive graph-based applications including finance domain applications. For example, in [24], GNN is employed for the stock prediction problem and the correlation between different stocks is formulated based on the shareholding graph. Nevertheless, due to restricted cross-shareholdings across public companies, this graph does not adequately illustrate the complicated interactions between stocks. Similarly, in [54], instead of forecasting stock prices, stock movement is forecasted by employing domain knowledge to build the stock relationships using the GNN and GRU models. They have validated their results on two stock indices of the Chinese

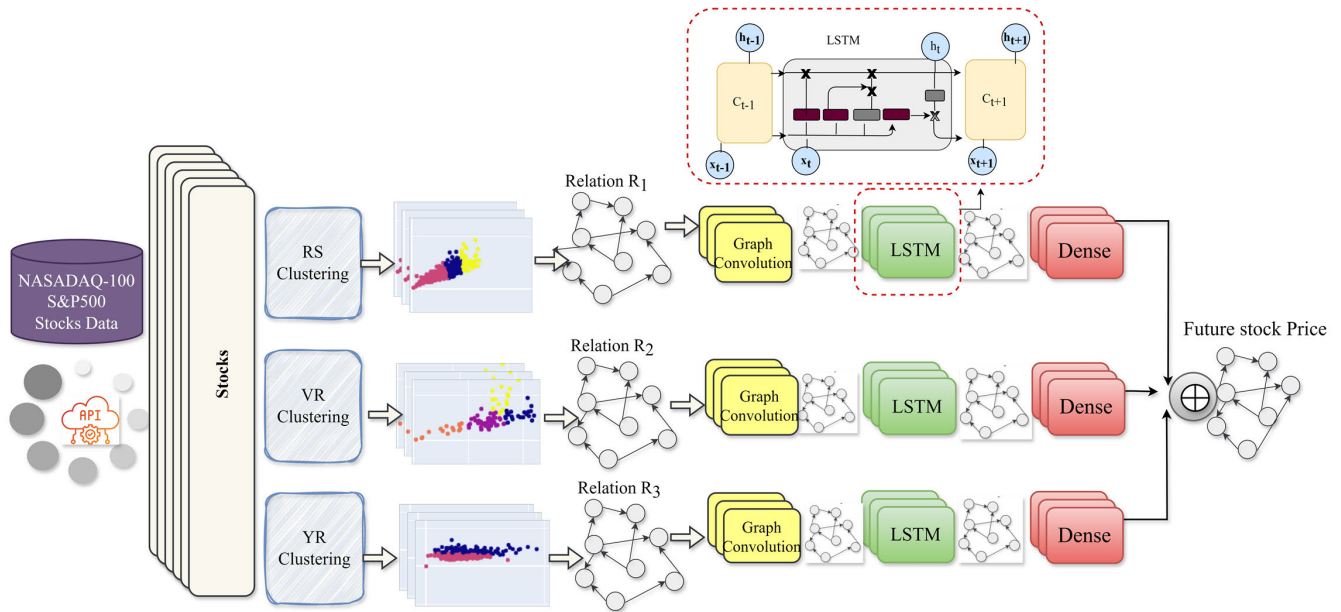


FIGURE 4. Pictorial overview of the proposed MCG Model for stock price forecasting.

market, and it is observed that their proposed method outperforms the existing methods. Following on, in [55], a temporal and heterogeneous GNN model is suggested for time series prediction of stock prices. More precisely, their graph model automatically learns the relations among stock prices. A large set of experiments are conducted on China and US stock markets to validate the results of their proposed methods, and the findings show the superiority of their approach. Another model named GCNET is also proposed in [56]. In this model, relationships among arbitrary sets of stocks in the form of a graph structure known as an influence network are created followed by employing a collection of predictive models to generate initial target values for different nodes of the graph symbolizing different stocks. The main aim of this method is to predict stock movement instead of explicit stock prices. The graph attention network is also exploited in [57] to anticipate stocks. To generate the graphs, they have utilized the database of Wikidata. Similarly, in [58], attentional graph neural networks are used to correlate social media content with corporate correlations. A multi-modality-graph neural network is also designed in [59], by involving historical data, news information, and events-related information, for forecasting price movements. They have utilized a two-phase attention technique for collaborative optimization, to guarantee model interpretation as well as enabling end users to access the significance of inner- and inter-modality sources. It is observed from existing studies, that GNN models are mostly applied for stock movement prediction instead of explicit stock price prediction, as there are only a few works in which stock prices task is undertaken. Secondly, the graphs are built on different kinds of relationships e.g. corporation relationships [24], however, an appropriate technique to build graphs is an ongoing area of research as well as the gap to be filled.

III. PROPOSED METHODOLOGY

This section provides an in-depth explanation of how the proposed model works. More specifically, the suggested model MCG-Network (Multi-cluster graph network) includes clustering techniques for building various graphs, such as RS (Returns-Sharpe Ratio), VR (Volatility-Return), and YR (Yearly Variance and Returns) clustering. Such clusters are used to generate graphs, which are then fed into graph convolution layers, followed by an LSTM model on each node of the graph model for each company/stock. Figure 4 shows the pictorial representation of the proposed work.

A. BACKGROUND AND PROBLEM FORMULATION

Recent research is centered on generalizing neural networks to graph-structured data. The enduring popularity of CNN inspired researchers to redefine convolution as graph convolution, a type of convolution using graph data. In the recent past, its application has also advanced the field of finance. Generally, stock price forecasting is categorized into two main themes of methodologies such as stock price prediction as well as stock movement forecasting. More specifically, in stock price forecasting, the precise value of the following day's stock price is anticipated, but in stock trend or movement prediction, the future is just anticipated to be in either an upward or downward trend, as a binary classification problem. Predicting exact prices in the stock market is challenging due to its complexity and stochasticity. However, predicting stock movement is somewhat feasible. Hence most of the studies focused on stock movement prediction rather than stock price. The major objective of stock price prediction is to predict the future price of a target stock from the collection of stocks with different future horizons using historical prices of particular stocks. Mathematically, this is described

as follows:

$$y_d^s = f[x_t^s] = f[x_{t-P}^s, \dots, x_{t-1}^s] \quad (1)$$

In the above equation (1), y_d^s is the next d day future price of stock s , where P is the lag size x_t^s is the stock's historical closing prices at previous time steps. However, this approach ignores the complicated relationships between stocks. Hence, this study employs graphs constructed using clustering among stocks to symbolize stock correlations and analyze their impact on stock forecasting. More explicitly, the graph $G = (V, E)$ comprises vertices and edges in which stocks are represented on vertices also called nodes while edges show the relationship among stocks. Suppose if there are m stocks denoted as $Stocks = \{s_1, s_2, s_3, \dots, s_m\}$ the nodes or vertices are represented as $V = \{v_{s1}, v_{s2}, v_{s3}, \dots, v_{sm}\}$. Every edge denoted as $e_{i,j}$ connects two nodes in a graph denoted as v_{si} and v_{sj} where $i \neq j$. These edges are weighted to symbolize the weight of the particular edge $e_{i,j}$, the $w_{i,j}$ notation is utilized. In the proposed MCG-Network, this $w_{i,j}$ has a value of 1 between v_{si} and v_{sj} , if both s_i and s_j belong to the same cluster other wise $w_{i,j} = 0$ indicating no edge between v_{si} and v_{sj} . The clustered stocks are employed to create a graph, with each stock symbolized as a node having an edge created between nodes that fall under the identical cluster. This graph topology enables the GNN to acquire knowledge from the associations among stocks with comparable financial features. The GNN then analyzes this knowledge to forecast future market patterns, drawing on individual stock data and prevalent patterns within every cluster. In this research, stocks are clustered using several techniques, and each strategy results in a graph network that is used as input to the Graph-based LSTM model.

B. STOCK CLUSTERING

Clustering is a foundational unsupervised machine learning approach that groups similar data points collectively according to common features as well as attributes. In the context of stock clustering, this approach is used on historical stock market data to uncover associations and similarities between stocks. For instance, in [60], the authors have employed a clustering algorithm for the construction of portfolios by choosing clusters of stocks having good returns. Likewise, [61] investigates clustering prices historical data to develop a stock trading recommender system. For efficient portfolio management, stock clustering is carried out [62] to identify stock clusters having strong trends as well as momentum that might lead to superior performance in a short period of time. Contrary to that, the existing graph neural networks-based stock movement prediction or stock price prediction models exploit the relationship among stocks based on some correlation, shareholdings, industry, and sector-base relationship in which threshold has been defined and decided to include an edge between stocks or not. The proposed clustering technique provides a more detailed and adaptable depiction of stock relationships. For example, the direct sector-based relationship might overlook some relations resulting in a

sparse adjacency matrix since stocks from different sectors may have comparable financial features, opening up possible investment possibilities. As a result, clustering is a better strategy for grouping stocks and determining their correlations. In this study, the graph networks are built using different clustering strategies. A pictorial overview of clustering stocks for building graph models is depicted in Figure 5.

C. K-MEANS CLUSTERING

K-Means clustering is one of the popular clustering algorithms employed for partitioning data into meaningful clusters K . It usually starts by randomly initializing centroids and these centroid points are regarding data points representing each cluster. Following on, in successive iterations, it continuously updates centroid positions and assigns data points to different clusters until there is no change in centroids. K-means clustering aims to reduce the within-cluster sum of squared distances (WCSS), which is referred to as inertia or distortion. WCSS is calculated as the total of the squared Euclidean distances among each data point (here the stocks are data points) as well as its centroid. The Euclidean distance between two stocks is defined as follows:

$$d(p, q) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2} \quad (2)$$

In the above equation (2), p and q are the stock's data points on which clustering has been performed.

1) RS-CLUSTERING

In RS clustering two metrics including annual returns and Sharpe ratio are computed for each stock present in the NASDAQ-100 and S&P500 Index and based on these metrics the stocks are grouped into clusters using the K -mean algorithm. More precisely, the annual return is the amount earned at the close of a 365-day calendar year. Mathematically, it is as follows:

$$r = (1 + R)^{\frac{365}{t}} - 1 \quad (3)$$

In the previous equation (3), t represents the number of trading days. Likewise, the Sharpe ratio also provides insights about risk-adjusted relative returns, and it is computed as follows:

$$S_T = \frac{\text{mean}(R_t) - r_f}{\text{std}(R_t)} \quad (4)$$

where

$$R_t = \frac{v_t - v_{t-1}}{v_{t-1}}$$

In the above equation (4), r_f indicates a risk-free rate. Once these indicators have been calculated for each stock, the K -means clustering method is used to divide the stocks into clusters. This algorithm classifies stocks by realizing their patterns in yearly returns and Sharpe ratios, then grouping them into categories based on these metrics. More precisely, if two stocks, namely 'AAPL' and 'CSCO', have the same

performance profile in terms of Sharpe ratio and annual returns, they belong to the same cluster, and the nodes in the graph symbolizing these stocks are connected by an edge.

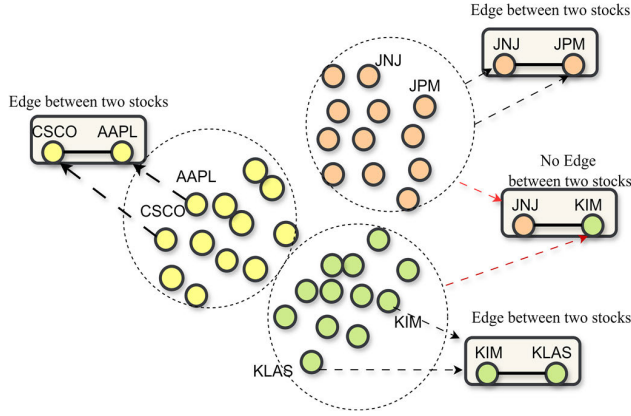


FIGURE 5. Pictorial overview of clustering stocks for building graph models.

2) VR CLUSTERING

In this method, annual volatility and returns are used to identify the clustered stocks. Annual volatility is used for determining the risk of a portfolio or investment by computing how much the price will fluctuate throughout an interval of time. It is defined as.

$$\sigma_a = \sqrt{\frac{\sum_{i=1}^n (r_i - \bar{r})^2}{n-1}} \quad (5)$$

In the above equation (5), i indicates the year for which annual returns are determined symbolized by r_i while $\bar{r}$ is showing the mean annual return along with the total number of years represented by n . Similar to the previous technique, the yearly volatility and returns for all stocks in the NASDAQ-100 and S&P500 are obtained, and then k-means clustering is used to group stocks into clusters. In this method, if two stocks show the same pattern over time in terms of yearly volatility and returns, they belong to the same cluster and are connected by an edge on the graph.

3) YR CLUSTERING

In this method, yearly variance and returns are computed to exploit the relationships among stocks using clustering. A stock's historical variance is the disparity among its returns over various time frames against the mean return. A stock with reduced volatility produces returns that are nearer to the mean. Just like the last two methods, yearly variance and returns are inputted as data to the K -means clustering in this strategy and stocks are clustered based on their yearly variance and returns they made. The stocks are grouped into a single cluster if they have identical values of yearly variance and returns and they are connected by an edge in the GNN model.

D. GRAPH CONVOLUTION LAYERS

Graph Convolutional Network (GCN) is a paradigm for learning representations in graphs based on graph signals theory. GCN may be used explicitly on data with a graph structure to obtain beneficial information for every node by combining details from adjacent nodes at depth d . In the recent past, different researchers have aimed to derive high-level representations of graph structures from processed data. Based on the theory of convolution on images, the convolution on graphs has been first formalized in the spectral domain. The decomposition of graph Laplacian matrix after normalization is accomplished as suppose the matrix is $L = I_N - D^{-\frac{1}{2}}AD^{-\frac{1}{2}} \in \mathbb{R}^{N \times N}$ and after decomposition $L = U\Lambda U^T$. In this decomposition, the graph Fourier basis U is acquired comprising the topological details of the graph, and D denotes the degree matrix of A , while eigenvalues are provided as Λ with U as the eigenvector matrix. The $x \in \mathbb{R}^N$ indicating the graph signal is initially converted into the spectral domain with the assistance of graph Fourier U followed by filtering kernel Θ and in the last, it is translated back into inverse graph Fourier basis denoted as U^T to acquire the outcome of the convolution operation as follows:

$$y = \Theta * \mathcal{G}x = U\Theta U^T x \quad (6)$$

In the above equation (6), $* \mathcal{G}$ is the graph convolution operation. A kernel Θ having N weights consider all nodes in the network for convolution. The kernel value Θ is limited with respect to the polynomial of eigenvalue to acquire k -hop convolution:

$$y = \Theta(\Lambda) * \mathcal{G}x = U \left(\sum_{k=0}^{K-1} \theta_k \Lambda^k \right) U^T x = \sum_{k=0}^{K-1} \theta_k L^k x \quad (7)$$

In the above equation (7), θ belongs to $\mathbb{R}^K$ and is referred to as vectors of polynomial coefficients while K is the size of the filter. This is basically the spectral graph convolution operated on the input data of graph x . This operation may involve intensive calculations. To address this, spectral kernels are approximated in the form of Chebyshev polynomials and restrict the layer-by-layer convolution process to $K = 1$ to mitigate the issues of over-fitting on local neighbor topologies for networks with massive node degree ranges hence creating the graph convolution networks. GCN transforms Equation (9) into a fully connected layer that includes a built-in convolution kernel, as seen in the equation (9):

$$H^{l+1} = \text{Relu}(\hat{A}H^l W^l) \quad (8)$$

In the above equation, $\hat{A} = \hat{D}^{-\frac{1}{2}}\hat{A}\hat{D}^{-\frac{1}{2}}$ and $\hat{A} = A + I_N$ and $\hat{A}$ hold the adjacency matrix of the underlying graph, along with self-connections, while I_N shows the identity matrix and $\hat{D} = \sum_j \hat{A}_{i,j}$ with trainable parameters of the layer are indicated by W^l . Therefore, the above equation (8), becomes:

$$H^{l+1} = \rho \left(\sum_{k=0}^{K-1} \theta_k L^k \right) H^l W^l \quad (9)$$

where L indicates a graph reliant on stock relationship while $H^l \in \mathbb{R}^{N \times F}$ is the input provided at l -th layer, and $\rho(\cdot)$ represents the activation function, such as ReLu. The first stage in GCN is to gather knowledge about nearby stocks by multiplying the Laplacian as well as feature matrices. A dense layer is then incorporated to aggregate characteristics to provide abstract representations for every single stock.

E. GRAPH-NODES-BASED LSTM MODEL

Following the graph convolution layers, the information on each node is propagated through LSTM layers. LSTM stands for long short-term memory networks and they are one of the variants of recurrent neural networks (RNN). More precisely, LSTM is the extended version of RNN with a better ability for time-series forecasting. One of the potential differences between the two models is that LSTM is able to hold long-range time dependencies by involving special units or cells called memory cells. It also addresses the problem of vanishing gradients occurring in RNNs. The LSTM design comprises memory cells as well as three types of gates: input, forget, and output. These gates control the passage of information through and out of memory cells, allowing a network to dynamically keep and update data across many time steps. The purpose of the input gate is to determine which value should be employed to alter the memory while the forget gate determines which information needs to be forgotten from memory and the output is decided by the output gate. The output of forget is obtained using the following equation:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (10)$$

In the above equation (10), the W_f denotes the weights associated with the forget gate, b_f represents the bias term, while the input value of the stock price at the current time step is denoted by x_t , the previous output values are denoted by h_{t-1} and σ denotes sigmoid activation. Likewise, the output value of the stock price at the last time step and the current time step is provided as input to the input gate to compute the value of output as well as the candidate cell state.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (11)$$

$$\hat{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (12)$$

In the above equations (11)-(12), the W_i denotes the weights associated with the input gate, and the current cell is updated as follows:

$$C_t = (f_t * C_{t-1} + i_t * \hat{C}_t) \quad (13)$$

Following on, the value at the output gate is computed by taking h_{t-1} as well as the input x_t :

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (14)$$

Similar to previous gates, W_o and b_o denote the weights and biases associated with the output gate. The final result of the LSTM is derived by computing the outcome of the output gate as well as the current state of the cell, as shown below:

$$h_t = o_t * \tanh(C_t) \quad (15)$$

In this research, these LSTM layers are added over each graph node to predict future stock prices of stocks by considering the relation of a particular node (i.e. representing particular stock e.g. AAPL) of the graph with its corresponding nodes using graph convolution layers during training. Following on, dense layers are added with a linear activation to predict the future price of the stock. More precisely, following the graph convolution layers that have processed the incoming data, the spatial attributes of every node (which symbolizes a stock) are updated depending on its graph neighbors.

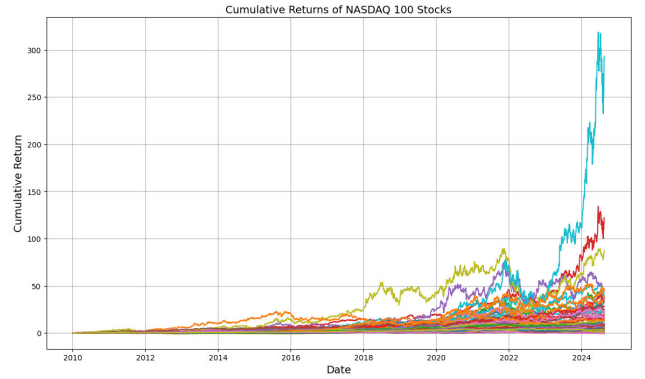


FIGURE 6. Cumulative returns of NASDAQ stocks.

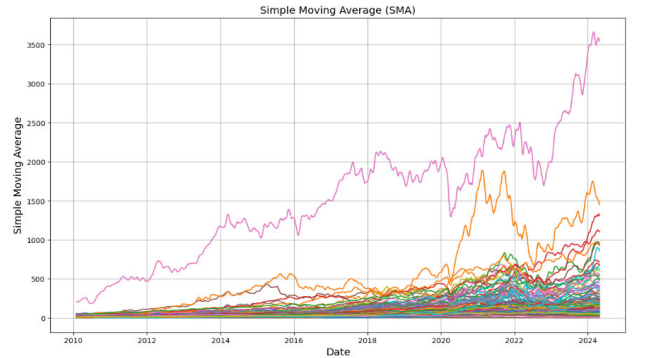


FIGURE 7. Simple Moving Average (SMA) of NASDAQ stocks.

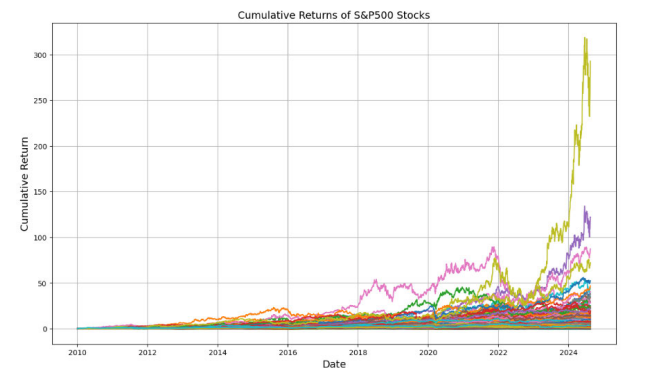


FIGURE 8. Cumulative returns of S&P500 stocks.

In addition to this, each node symbolizes stocks and also holds the temporal information of stock which is going to be reshaped to become the input sequence of the LSTM layer

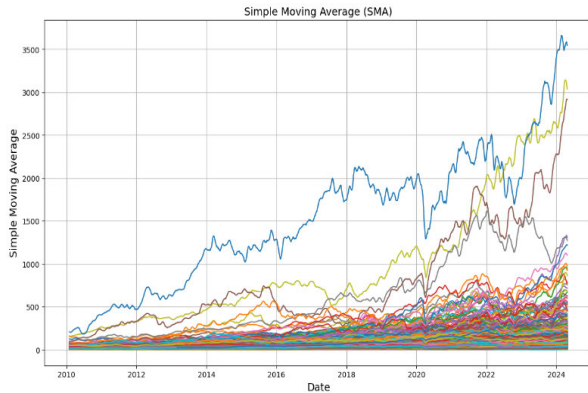


FIGURE 9. Simple Moving Average (SMA) of S&P500 stocks.

for each node. Since we employed multi-relation graphs, each graph with its own topological structure according to the adjacency matrix is processed separately using graph convolution layers LSTM cells. In the end, all the outputs are passed through fully connected layers. This amalgamated result incorporates information from all of the individual graph relations.

F. HYPERPARAMETERS AND TRAINING PLATFORMS

The proposed model's hyperparameters include 50 epochs an Adam weight optimizer with a learning rate of 0.01, a mean-squared error function, and a 64-batch size. Furthermore, the LSTM units are set to 64, whereas the total units in the final dense layer are 1 using a linear activation function. The hardware-software platforms include Google Collaboratory with the Free GPU package for training our model, and Python is used for implementation.

IV. EXPERIMENTS AND RESULTS

In this section, details of the dataset, the evaluation criteria used to evaluate the model, and the results of the proposed MCG model are presented.

A. DATA ACQUISITION

In this study, the daily stock price data of the NASDAQ 100 INDEX comprising 100 stocks and the S&P500 Index comprising 500 stocks were retrieved from Yahoo Finance API for the time span of 14 years i.e. 2010-01-01 to 2024-01-01. Figure 6 and Figure 7 show the cumulative returns and SMA of 100 NASDAQ stocks and Figures 8 and 9 show the cumulative returns and SMA of 500 S&P stocks. These Figures indicate that these stocks have an elevated degree of volatility. Furthermore, the implementation has been carried out in Python, and experiments are performed on Google Colab using an NVIDIA Tesla T4 GPU and 12GB RAM.

B. EVALUATION METRICS

The evaluation metrics employed to assess the performance of the model are given below:

1) MEAN ABSOLUTE ERROR (MAE)

MAE is one of the potential metrics used for the performance of regression tasks. It computes the absolute difference between actual and predicted stock price values. Moreover, it does not take a square of errors, which indicates that this metric assigns equal weight to all errors. Mathematically, it is defined in equation (16):

$$MAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n} \quad (16)$$

In the above equation (16), n denotes the total number of testing instances while y_i is the actual value and $\hat{y}_i$ is the predicted value from the MCG model.

2) MEAN SQUARED ERROR (MSE)

MSE is also one of the potential metrics used in the regression algorithms. It observes the mean squared difference between actual and anticipated values from the MCG model. The mean squared error is often called the mean squared deviation (MSD). Mathematically, it is defined in equation (17):

$$MSE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n} \quad (17)$$

In the above equation (17), n denotes the total number of testing instances while y_i is the actual value and $\hat{y}_i$ is the predicted value from the MCG model.

3) ROOT MEAN SQUARED ERROR (RMSE)

Root means square error (RMSE), also known as root mean square deviation, is one of the most often used methods for measuring how well the MCG model performs. It is computed by taking the square root of the mean squared error. Mathematically, it is defined in equation (18):

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (18)$$

In the above equation (18), n denotes the total number of testing instances while y_i is the actual value and $\hat{y}_i$ is the predicted value from the MCG model.

a: K-MEANS CLUSTERING (K-MEANS) RESULTS

To access the performance of the proposed MCG model we have first collected data from Yahoo finance API for both indices. MCG model's initial stage is the building of multi-relation graphs computed from three clustering strategies i.e. RS-based clustering, VR-based clustering, and YR-based clustering. To carry out clustering, different measures including Sharpe ratio, yearly variance, annual returns, and volatility are computed for each stock. Some partial graphs resulting from clustering strategies are depicted in Figure 10. Each node in the graph represents a particular stock while the edges show the relationship between two stocks. Initially, the Sharpe ratio, yearly variance, annual returns, and volatility are computed for initiating the clustering using k-means. The best number of clusters can be identified using

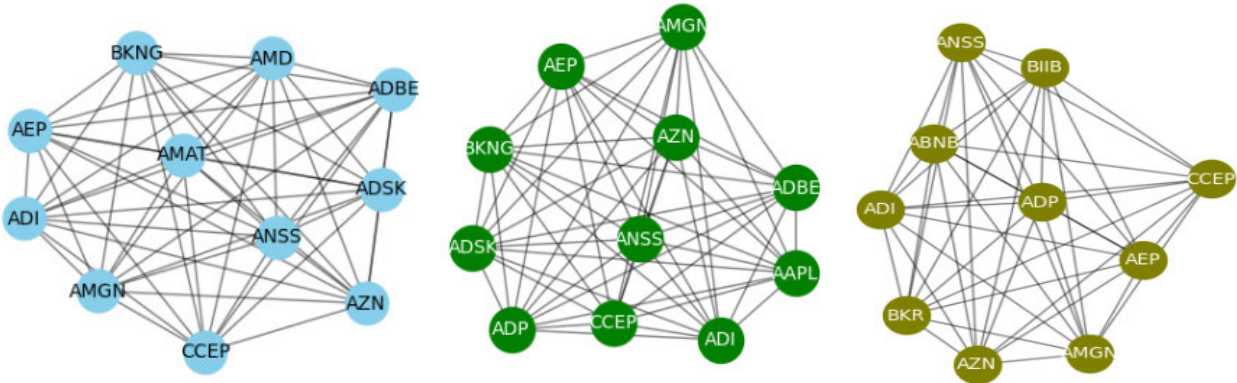


FIGURE 10. Pictorial representation of partial graphs resulting from different clustering strategies (a) RS Clustering. (b) VR Clustering (c) YR Clustering.

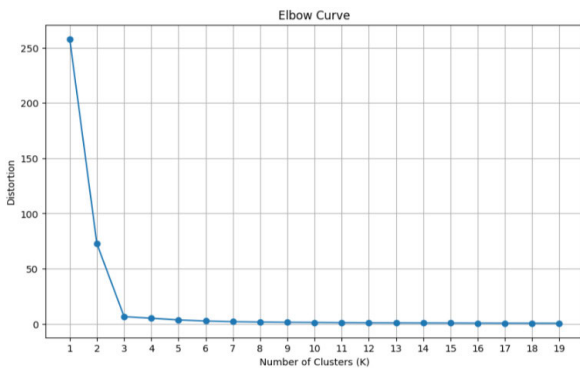


FIGURE 11. Elbow Curve of K-means clustering algorithm.

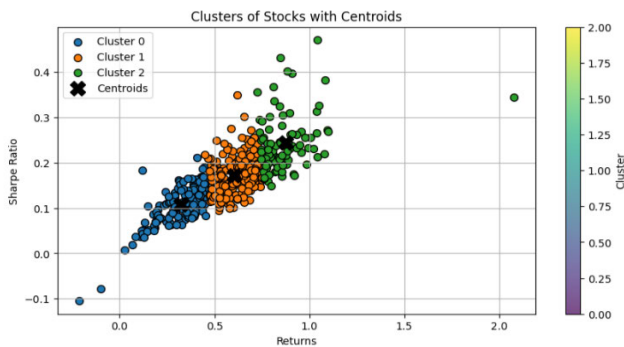


FIGURE 12. Clustering (RS) on the S&P500 Index comprising 500 companies.

the elbow approach. An elbow graph for RS-based clustering of S&P500 index stocks is depicted in Figure 11. It is obvious from Figure 11 that the optimal number of clusters is 3 for which distortion is minimal. Moreover, the clustering results are also depicted in Figure 12. In Figure 12, each data point represents the particular stock or company position based on the Sharpe ratio and annual returns. The black “x” denotes the centroid of each cluster. The total number of stocks and

companies is 500 in the S&P500 Index resulting in 500 data points on which clustering has been performed.

b: RESULTS OF THE PROPOSED MCG MODEL ON THE NASDAQ INDEX

To analyze the efficiency of the proposed MCG model, we first clustered stocks using three distinct strategies: RS (Returns-Sharp Ratio), VR (Volatility and Returns), and YR (Variance and Returns) clustering. More precisely, in the RS model, returns show a stock’s profitability during a certain time period, whereas the Sharpe ratio assesses a stock’s risk-adjusted return. Clustering by return and Sharpe ratio assists in identifying stocks that have comparable risk-return characteristics. Similarly, volatility shows the degree of change in a stock’s price over time, and clustering on this metric assist in determining or collecting stocks having identical volatility patterns. After clustering, the graphs on each strategy have been generated by creating neighbors of nodes in the graph depending upon their cluster. Three multi-relation graphs are given as input to the graph-based LSTM model to forecast future stock prices of each stock of NSADAQ 100 present on each node. The results of the proposed MCG model in terms of RMSE, MSE, and MAE are given in Tables 1, 2, and 3. More specifically, we have trained the model with different input sequences and future horizons. The input sequences are the prior time steps used to anticipate future prices, whereas the future horizon is the forecasting of future time steps, such as one day ahead, two days ahead, and so on. For instance, the lowest MSE, MAE, and RMSE error is observed with 1-time step’s input and the future horizon is about 0.01 0.08, and 0.11 respectively. The MCG models’ lowest numbers show their effectiveness in forecasting the future prices of individual stocks within a portfolio or index. This efficacy is particularly important since it indicates the models’ capacity to give precise forecasts, implying that they can capture the fundamental dynamics and trends in the stock market. By yielding lower error values, the MCG model shows a greater alignment between anticipated and actual

TABLE 1. Results of proposed MCG Model with one time steps input sequence On NASDAQ-100 index.

	RS Clustering	VR Clustering	YR Clustering	Future horizon=1	Future horizon=2	Future horizon=3
MAE	✓	✓	✓	0.081435505908	0.0250791	0.041482
MSE	✓	✓	✓	0.01416773548	0.1093299	0.145423
RMSE	✓	✓	✓	0.119028297	0.158364	0.20367208

TABLE 2. Results of proposed MCG Model with five-time steps input sequence On NASDAQ-100 index.

	RS Clustering	VR Clustering	YR Clustering	Future horizon=1	Future horizon=2	Future horizon=3
MAE	✓	✓	✓	0.082251961	0.110657808	0.1454267
MSE	✓	✓	✓	0.01474126	0.0260158831	0.042691120
RMSE	✓	✓	✓	0.1214136298	0.16129439	0.2066182

TABLE 3. Results of proposed MCG Model with ten time steps input sequence On NASDAQ-100 index.

	RS Clustering	VR Clustering	YR Clustering	Future horizon=1	Future horizon=2	Future horizon=3
MAE	✓	✓	✓	0.07938226	0.11456300	0.14122
MSE	✓	✓	✓	0.01387601	0.0272118	0.0411140601
RMSE	✓	✓	✓	0.1177964	0.164960079	0.2027660232

prices, increasing their dependability in guiding investment decisions. The experiments were also carried out with different future horizons and input sequence lengths, and it was found that increasing the future horizon caused the error to increase somewhat, showing that the model had difficulty forecasting more than one day in advance. The actual and predicted prices of some stocks of NSADAQ-100 are provided in Figure 13. It is observed from Figure 13 that the difference between actual and predicted is very minimal indicating the worth of the proposed method.

c: COMPARISON BETWEEN MULTI-CLUSTER RELATION AND SINGLE-CLUSTER RELATION GRAPH ON NSADAQ-100 INDEX

To validate the performance of the proposed multi-cluster relation in comparison with a single relation graph, a set of experiments has been conducted. This set of experiments shows how multi-relation clustering assists in improving the model performance. The results of the single-clustering graph are provided in Tables 4, 5, and 6. In this case, the model also shows low error values, but the error is high in comparison with multi-relation clustering, especially in the future time horizon of 1 day ahead. This demonstrates the effectiveness of our technique for understanding complicated relationships among stocks. By using clustering algorithms that look into multiple relationships or elements at the same time, the model is better able to reflect the data's intricate dependencies and connections. This comprehensive approach to clustering allows the model to detect more complex patterns and relationships among the stocks, improving its forecasts and

reducing errors. Figure 14 shows the actual and forecasted prices for various NSADAQ-100 stocks.

d: RESULTS OF THE PROPOSED MCG MODEL ON THE NASDAQ INDEX

In the second experimental setup, we focused on analyzing the efficacy of the suggested MCG model on the S&P 500 Index, which includes a comprehensive portfolio of 500 companies or stocks. The selection of this vast array of companies is important in our performance assessment, owing to its representation of a diversified and large dataset. This selection lets us dig into the most important aspects of performance validation, notably determining the scalability of the suggested MCG model. The S&P 500 Index functions as a reliable benchmark for assessing the versatility and efficacy of the MCG model in practical scenarios. Similar to prior settings, we first cluster the stocks of the S&P Index in optimal value of clusters using the Elbow method. Three different strategies are employed for clustering including RS, VR, and YR clustering. Each exploits a useful set of features among stocks to build multi-relation graphs among clusters.

More specifically, under the RS model, returns reflect a stock's profitability over a certain time period, but the Sharpe ratio measures a stock's risk-adjusted return. Clustering by return and Sharpe ratio helps locate stocks with similar risk-return characteristics. Similarly, volatility measures the degree of change in a stock's price over time, and clustering on these metrics aids in the identification or gathering of stocks with similar volatility patterns. Such clustering strategies are employed to build graphs, and these graphs

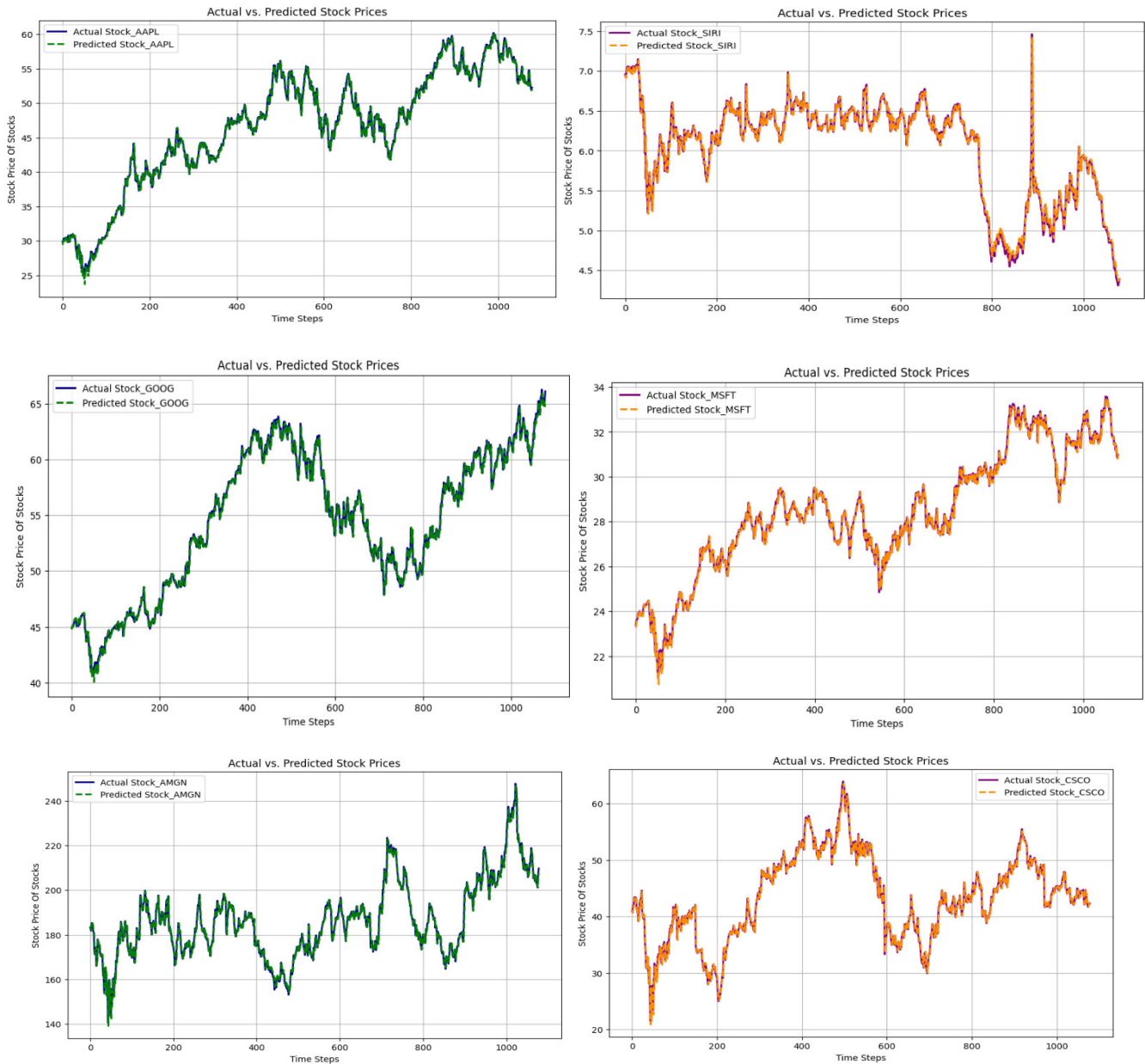


FIGURE 13. Actual and predicted future stock prices of some stocks of NASDAQ-100 Index represented at each node.

TABLE 4. Results of proposed MCG Model with one time steps input sequence using single relation graphs On NASDAQ-100 index.

	RS Clustering	VR Clustering	YR Clustering	Future horizon=1	Future horizon=2	Future horizon=3
MAE	✓	✗	✗	0.08522460	0.1240532	0.13131170
MSE	✓	✗	✗	0.0146623075	0.03055319	0.037027688
RMSE	✓	✗	✗	0.12108801567	0.174794731	0.1924258008

are provided as an input LSTM-based graph model. Following on, LSTM is utilized to forecast future prices of each stock that are represented as nodes of graphs. The results of the S&P500 Index with different input sequence lengths

and future horizons are provided in Tables 7, 8, and 9. It is observed from these results that the lowest error value of the proposed MCG on the S&P500 Index in terms of MAE, MSE, and RMSE is about 0.08, 0.01, and 0.12 which is very much

TABLE 5. Results of proposed MCG Model with FIVE time steps input sequence using single relation graphs On NASDAQ-100 index.

	RS Clustering	VR Clustering	YR Clustering	Future horizon=1	Future horizon=2	Future horizon=3
MAE	✗	✓	✗	0.0759549987	0.1171694563083	0.1342567391
MSE	✗	✓	✗	0.01307246100	0.02707796266	0.037604914
RMSE	✗	✓	✗	0.1143348634	0.164553829	0.1939198651

TABLE 6. Results of proposed MCG Model with TEN time steps input sequence using single relation graphs On NASDAQ-100 index.

	RS Clustering	VR Clustering	YR Clustering	Future horizon=1	Future horizon=2	Future horizon=3
MAE	✗	✗	✓	0.0857786012	0.1247286729	0.1427972
MSE	✗	✗	✓	0.01500908	0.031009867	0.0405278206
RMSE	✗	✗	✓	0.122511577	0.176096187	0.201315226

less. Similarly, the proposed MCG model also performs well with 2 and 3-time steps ahead forecasting. The error values in terms of MAE, MSE, and RMSE for the future horizon window of three are about 0.14, 0.04, and 0.209. This is particularly noteworthy given that the model is tasked with predicting three-time steps into the future, which inherently introduces additional complexity and uncertainty into the forecasting process. Furthermore, it is critical to emphasize the influence of input sequence length on predicting accuracy. We found that as the length of the input sequences increased, correspondingly increased the error values. This finding is supported by the fact that lengthier input sequences typically contain more prior information, possibly bringing a larger amount of noise into the model. Given this, the rise in error values is just minimal, indicating that the model is flexible to the inclusion of new historical data. Similarly, the same impact has been observed, when we increase the input size of 10 previous time steps information to forecast the next 1 day, 2 days, and 3 days ahead forecasting of future prices of each stock. The error values in terms of MSE, MAE, and RMSE in this case with 1-day, 2-day, and 3-day ahead are higher. For instance, the errors are 0.18, 0.06, and 0.24 which is higher in comparison with the case when we consider the input sequence length of 1 day. This indicates that future prices are much more dependent on recent previous days rather than too much old information and history. Hence, it logically concluded that the proposed model shows good results in terms of error values over such a large set of companies. A comparative analysis for the single and multi-relation MCG model on the S&P500 Index is also shown in Figure 17.

e: COMPARISON BETWEEN MULTI-CLUSTER RELATION AND SINGLE-CLUSTER RELATION GRAPH ON THE S&P500 INDEX

Similar to the previous experimental setting, we have validated the impact of single-cluster relation and multi-cluster relation graphs on the S&P500 Index. As stated previously, the S&P500 Index provides an extensive array of companies from many various industries and sectors, giving a complete

and comprehensive testbed for evaluating the model's abilities across a variety of market situations and organizational frameworks. In this setup, clustering is performed with each strategy, and a single graph resulting from each strategy is then built to provide input to the graph-based LSTM model.

The results of each clustering strategy are provided in Tables 10, 11, and 12. The input time step considered to forecast future 1-day, 2-day, and 3-day prices is set to 1. The results in terms of MAE, MSE, and RMSE for the future horizon window of 1 are about 0.106, 0.022, and 0.150. Likewise, the error values in terms of MAE, MSE, and RMSE with a future horizon window of 2 are about 0.147, 0.039, and 0.19, while with a future horizon window of 3 are about 0.149, 0.0450, and 0.212 respectively. It is observed that with a single clustering graph, the results are a little bit lower indicating the multiple-relation graph resulting in three clustering strategies is more important for the MCG model to forecast future prices more efficiently. More explicitly, the inference made from this observation emphasizes the importance of the multiple-relation graph paradigm. By adding several clustering algorithms, the model acquires a more thorough grasp of the fundamental connections among stocks and relationships in the dataset.

f: COMPARATIVE ANALYSIS WITH SOME BASELINE METHODS

To comprehensively examine the performance of the proposed MCG model, we contrasted its findings with two baseline strategies: a basic LSTM model and a naïve martingale model (i.e. using the stock price on day t forecast for stock price on day $t + h$, where h can be any horizon). The assessment metrics employed in the present study were RMSE (Root Mean Squared Error), MSE (Mean Squared Error), and MAE (Mean Absolute Error), and the findings are shown in Tables 13 and 14. Table 12 illustrates the performance of stocks from the NASDAQ 100 Index, whereas Table 14 focuses on stocks from the S&P 500 Index. The results show that the proposed MCG-based model beats both

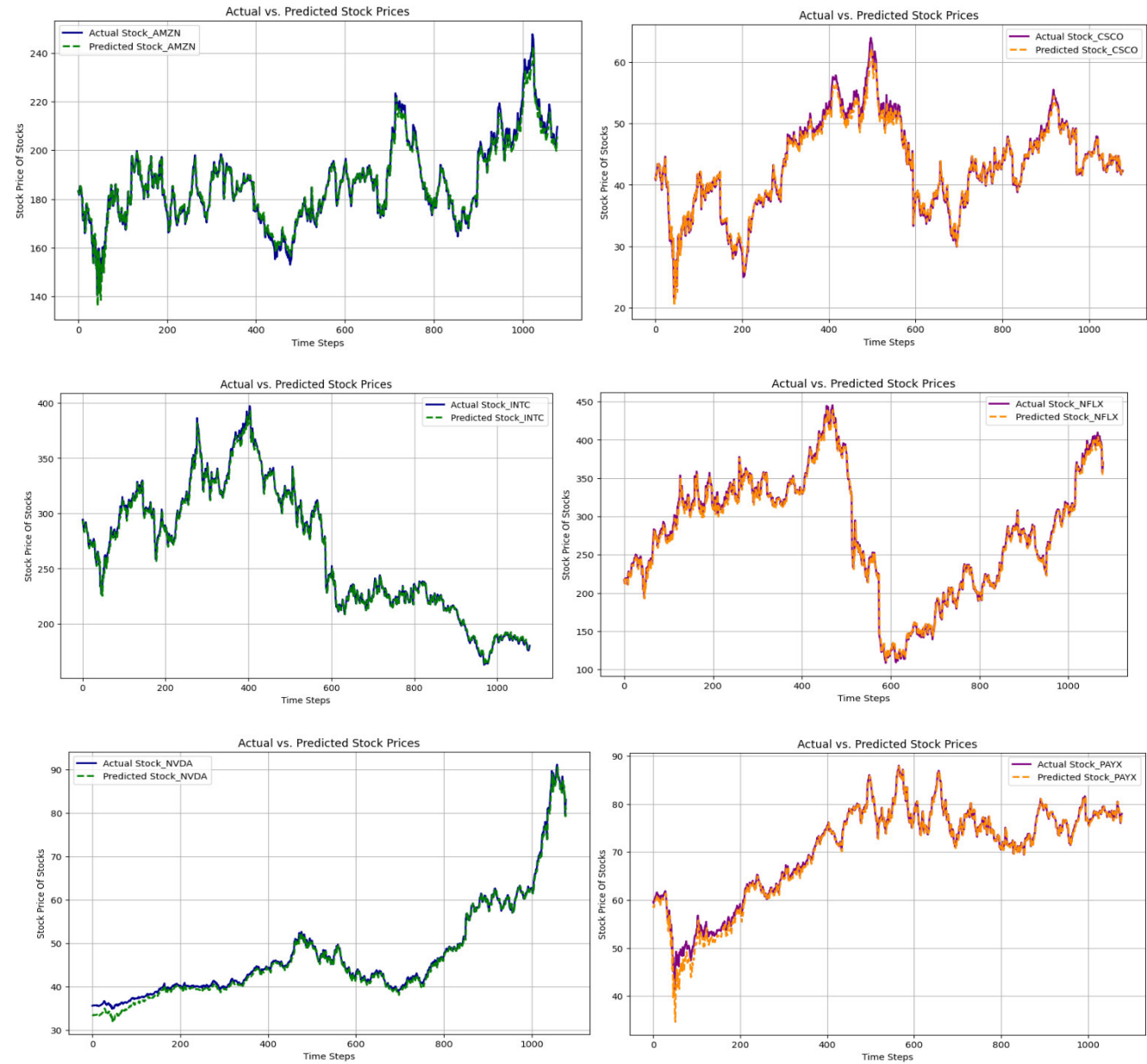


FIGURE 14. Actual and predicted future stock prices of some stocks of NASDAQ-100 Index represented at each node using single-relation clustering.

TABLE 7. Results of proposed MCG Model with one time steps input sequence on S&P500 index.

	RS Clustering	VR Clustering	YR Clustering	Future horizon=1	Future horizon=2	Future horizon=3
MAE	✓	✓	✓	0.084117149454	0.11944653	0.147251858
MSE	✓	✓	✓	0.0148173	0.028550768	0.04386572
RMSE	✓	✓	✓	0.1217264	0.168969	0.2094414506

baseline techniques on every criterion. Furthermore, scatter plots for some stocks, as shown in Figure 18, clearly demonstrate these performance disparities. For instance, in Figures 18(a) and 18(b), the LSTM model predicts higher

deviations from real stock prices than the GNN model (i.e. MCG), demonstrating that the GNN is substantially more exact. Overall, the findings show that the proposed MCG model outperforms standard baselines, especially in

TABLE 8. Results of the proposed MCG Model with Five time steps input sequence on the S&P500 index.

	RS Clustering	VR Clustering	YR Clustering	Future horizon=1	Future horizon=2	Future horizon=3
MAE	✓	✓	✓	0.0960103	0.12967754	0.148985164
MSE	✓	✓	✓	0.01975751	0.034867222	0.046168123
RMSE	✓	✓	✓	0.1405614279	0.186727668329	0.2148676872

TABLE 9. Results of proposed MCG model with TEN time steps input sequence on S&P500 index.

	RS Clustering	VR Clustering	YR Clustering	Future horizon=1	Future horizon=2	Future horizon=3
MAE	✓	✓	✓	0.1086928	0.145637819	0.1838945
MSE	✓	✓	✓	0.024029	0.041882	0.0623241
RMSE	✓	✓	✓	0.1550136	0.20465134	0.2496480

TABLE 10. Results of proposed MCG Model with ONE time steps input sequence using single relation graphs on S&P500 index.

	RS Clustering	VR Clustering	YR Clustering	Future horizon=1	Future horizon=2	Future horizon=3
MAE	✓	✓	✓	0.1063314	0.14753028	0.149766
MSE	✓	✓	✓	0.02256540	0.039298359	0.04505654
RMSE	✓	✓	✓	0.15021784	0.198238137	0.2122652

TABLE 11. Results of proposed MCG Model with Five time steps input sequence using single relation graphs on S&P500 index.

	RS Clustering	VR Clustering	YR Clustering	Future horizon=1	Future horizon=2	Future horizon=3
MAE	✓	✓	✓	0.10971593	0.150149501	0.16553409
MSE	✓	✓	✓	0.025354547	0.04226983	0.05534910
RMSE	✓	✓	✓	0.159231111	0.2055962	0.23526389

TABLE 12. Results of proposed MCG Model with TEN time steps input sequence using single relation graphs on S&P500 Index.

	RS Clustering	VR Clustering	YR Clustering	Future horizon=1	Future horizon=2	Future horizon=3
MAE	✓	✓	✓	0.1301259803	0.14072726	0.183725
MSE	✓	✓	✓	0.029351	0.03888670	0.06274018
RMSE	✓	✓	✓	0.17132382	0.1971971	0.25047

complicated financial forecasting situations. However, when comparing the three models, the proposed MCG outperforms the LSTM with larger margins model as well as the naïve martingale model with small margins, as shown in Figure 18 and Tables 13 and 14, respectively.

g: DISCUSSIONS AND IMPLICATIONS

Stock price forecasting is one of the hot research topics among researchers in the finance domain. Currently, the Artificial Intelligence (AI) domain is going to be much more advanced and advanced by spanning its applications in almost every field. As time passes, more sophisticated algorithms have been designed to mitigate and graph the intricacies of

the complex nature of data. Traditional stock price forecasting models, either machine learning and deep learning models or either stock movement/trend prediction or explicit stock predictions have been investigated and suggested by different researchers. However, in most of the methods, each stock is viewed as an independent entity while predicting its future price. This is because the primary premise and theory underlying these approaches is that stocks are independent of one another. These approaches ignore the cross-impact of stocks over time, which might have a dynamic effect on stock prices [21]. Nevertheless, with the advent of graph neural networks, researchers attempted to exploit its application in the finance domain to build graph networks between stocks. In this realm

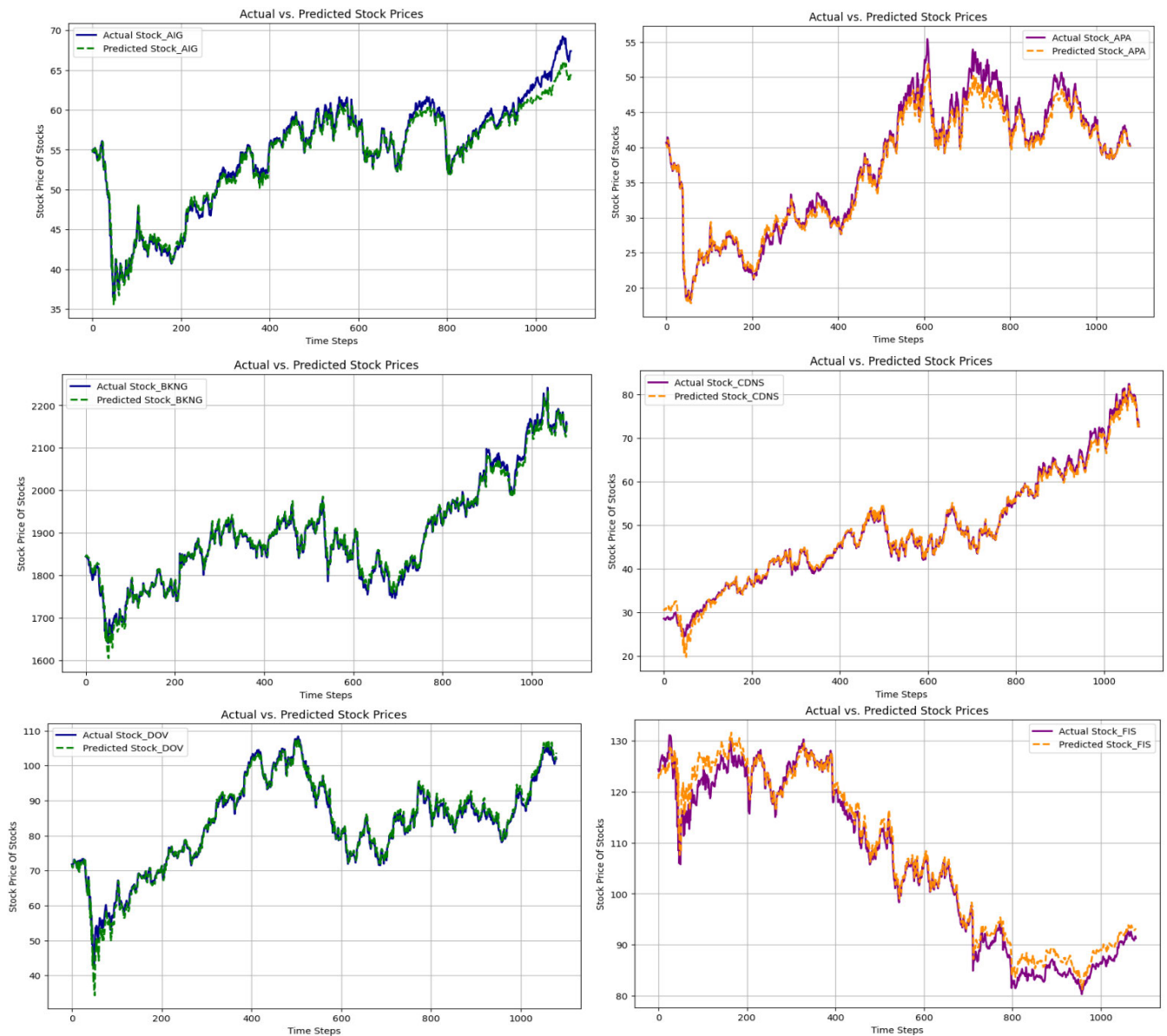


FIGURE 15. Actual and predicted future stock prices of some stocks of the S&P500 Index represented at each node using single-relation clustering.

of research studies, most of the work has been done for stock movement prediction using graph neural networks. More precisely, in this stock movement prediction, the stock prediction problem is conceptualized as a classification problem for which machine learning algorithms have been designed. The primary goal of this task is to anticipate the stock trend, i.e. whether the stock will be up or down tomorrow. This problem is simple rather than forecasting the exact future price of the model. Moreover, the relationship among stocks in existing GNN-based methods is modeled based on some heuristics such as if both stocks belong to the same sector, or if both stocks appear simultaneously in particular news headlines. Although these heuristics are somehow an approach to building graphs by exploiting relationships among stocks

these heuristics are not very accurate. For example, it may be the case, that two stocks do not belong to the same sector but exhibit the same volatility patterns or co-movements because of external variables including macroeconomic or sentiment variables. To handle such limitations, this research study proposed an MCG (Multi-Clustering Graph) model which is built by exploiting different relationships among stocks using three novel clustering approaches. These clustering approaches consider different metrics i.e. Sharpe ratio, yearly variance, annual returns, and annual volatility to cluster stocks, and after clustering the resulting graphs referred to as multi-relation graphs are propagated to LSTM-based graph convolution neural networks for forecasting of stock prices of each stock present at graph nodes. The experiments on two

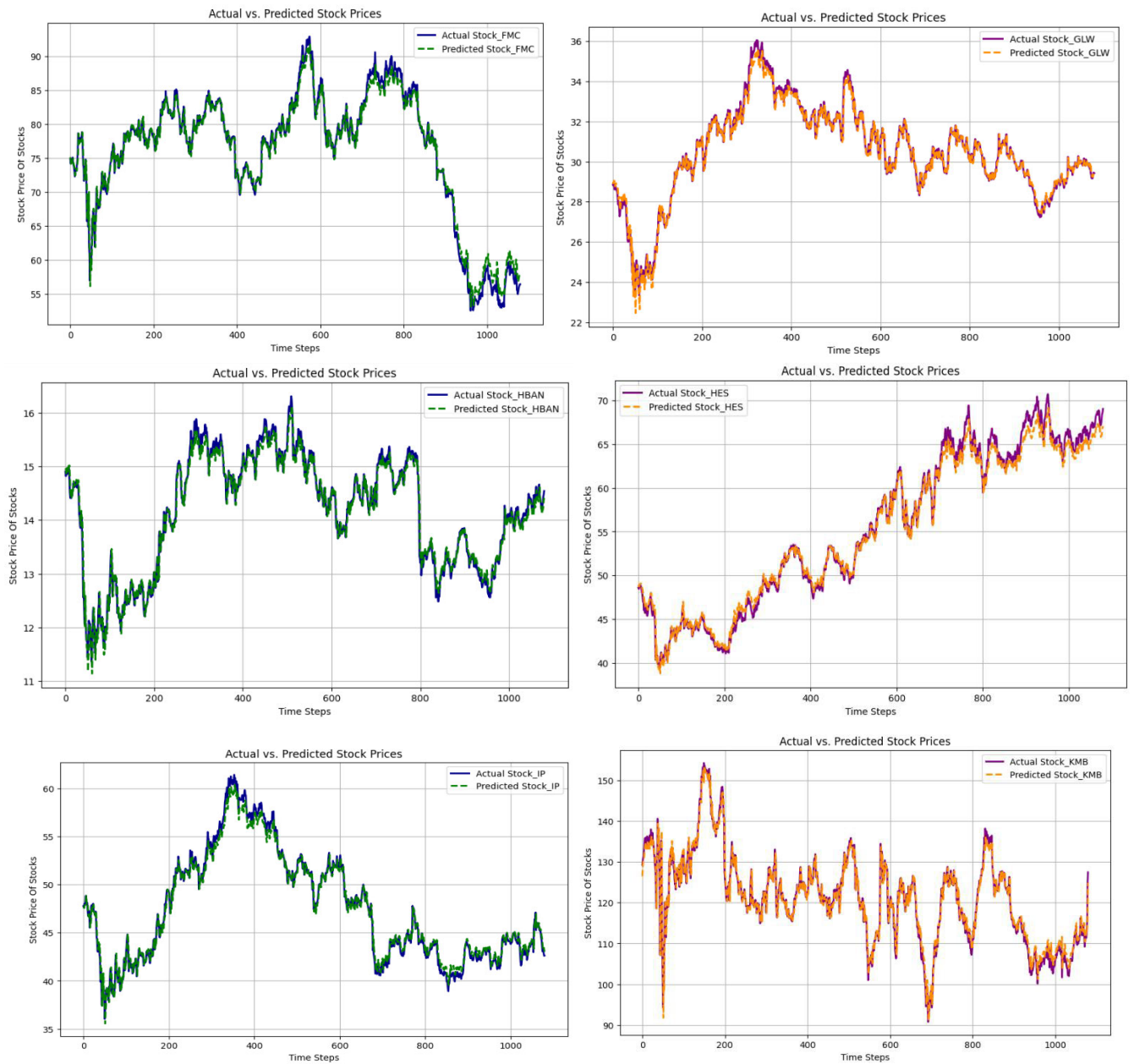


FIGURE 16. Actual and predicted future stock prices of some stocks of the S&P500 index represented at each node using single-relation clustering.

indices, namely NSDAQ-100 and the S&P500 Index show that the proposed MCG shows good results even in stocks having a higher level of volatility. The major implication of the proposed MCG model is its capability to exploit complex relationships among stocks in a more data-driven manner. GNNs can extract complicated patterns and relationships that would be difficult to detect when evaluating individual stocks using traditional machine learning methods. Moreover, in data-driven clustering-based methods, the hidden dependencies and patterns are more easily exploited to construct graphs which is not as apparent in the standard heuristic or threshold-criteria-based techniques. The experiments on

large-scale datasets, especially the S&P500 indexes comprising 500 companies demonstrate that the proposed model is scalable to real-world large datasets and environments. This index contains a diverse set of stocks and companies which is more related to the practical scenario to validate whether the proposed model is able to be deployed in real-world scenarios. In addition, graph models assist investors in understanding how interconnected stocks influence their stock price predictions. For example, a graph model may show how a quick decline in one stock affects another stock in an identical industry, assisting with handling risks and diversification of portfolios. In real-world scenarios, these

TABLE 13. Comparative analysis of the proposed model and baseline model on some stocks of the NASDAQ100 index.

Stock Symbol	RMSE	MAE	MSE	Model
SIRI	0.0238	0.0166	0.0006	LSTM
SIRI	0.0219	0.0147	0.0004	GNN
SIRI	0.0313	0.0188	0.0010	Naive Martingale
MRVL	0.0227	0.0166	0.0005	LSTM
MRVL	0.0210	0.0147	0.0004	GNN
MRVL	0.0237	0.0166	0.0006	Naive Martingale
NVDA	0.0694	0.0305	0.0048	LSTM
NVDA	0.0106	0.0064	0.0001	GNN
NVDA	0.0109	0.0066	0.00001	Naive Martingale
CDNS	0.0365	0.0333	0.0013	LSTM
CDNS	0.0118	0.0088	0.0001	GNN
CDNS	0.0133	0.0098	0.0002	Naive Martingale
BKNG	0.0196	0.0141	0.0004	LSTM
BKNG	0.0142	0.0105	0.0002	GNN
BKNG	0.0187	0.0136	0.0004	Naive Martingale

TABLE 14. Comparative analysis of the proposed model and baseline model on some stocks of the S&P 500 index.

Stock Symbol	RMSE	MAE	MSE	Model
ADP	0.0633	0.0542	0.0040	LSTM
ADP	0.0280	0.0247	0.0008	GNN
ADP	0.0209	0.0145	0.0004	Naive Martingale
ABT	0.0503	0.0447	0.0025	LSTM
ABT	0.0144	0.0105	0.0002	GNN
ABT	0.0219	0.0160	0.0005	Naive Martingale
ZION	0.0501	0.0348	0.0025	LSTM
ZION	0.0213	0.0158	0.0005	GNN
ZION	0.0237	0.0176	0.0006	Naive Martingale
ZBH	0.0263	0.0193	0.0007	LSTM
ZBH	0.0193	0.0135	0.0004	GNN
ZBH	0.0259	0.0179	0.0007	Naive Martingale
NFLX	0.0355	0.0267	0.0013(590.8350)	LSTM
NFLX	0.0176	0.0116	0.0003	GNN
NFLX	0.0299	0.0151	0.0005	Naive Martingale

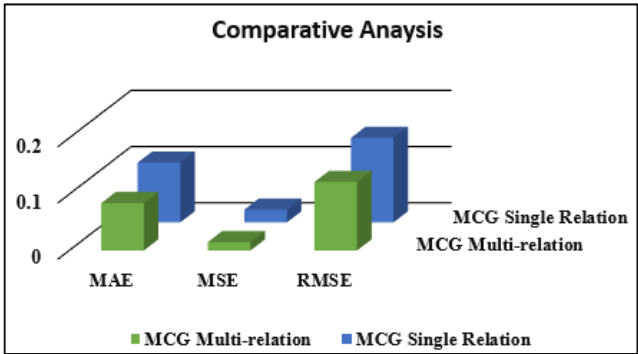


FIGURE 17. Comparative analysis of MCG multi and single relation graphs on S&P500 index stocks.

forecasting models assist investors in making stock trading decisions of buying and selling of shares. These systems also eliminate emotional bias while making decisions, since the

investor and traders usually face greed and fear because of fluctuations in stock prices [63].

Other than positive implications, there also exist some potential gaps that need to be improved and can be considered as a future research direction. Currently, the stock relations graphs are not adaptable e.g. in the future stock relations might be updated over time. To cope with this, more advanced methodologies such as reinforcement learning can be designed to update the relation graphs over time with the evolving conditions of the stock market. Furthermore, exploiting different data and variables e.g. supply chain relationships, news sentiments, etc., to determine more accurate stock-to-stock relationships for building graphs is also a potential future research direction. Various clustering techniques, such as DBSCAN (Density-Based Spatial Clustering of Applications with Noise) and Gaussian Mixture Models (GMMs), can be examined by employing clustering variables other than conventional metrics including returns and

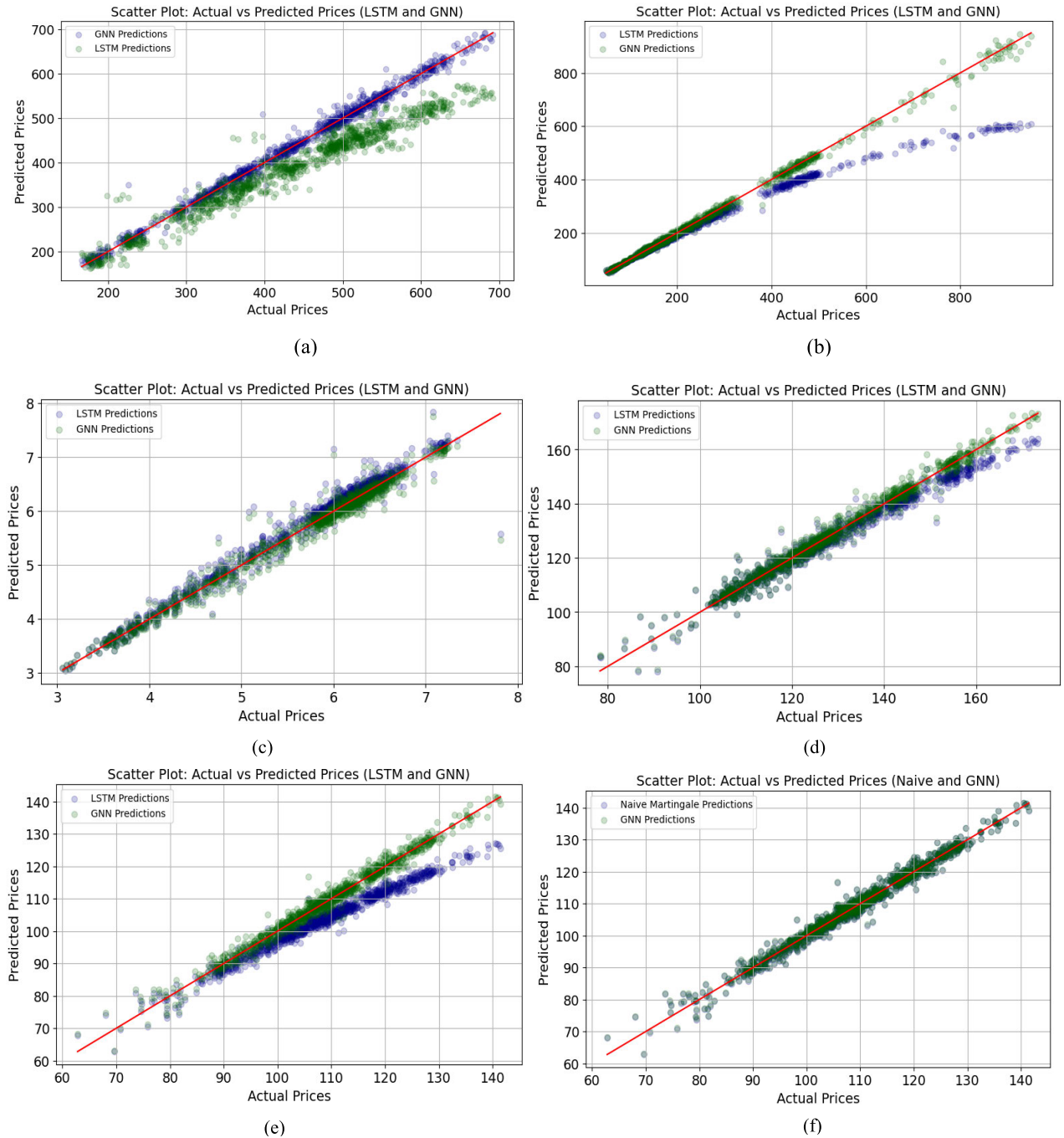


FIGURE 18. Scatter-plots of actual and predicted stock prices (a) NFLX Stock-GNN-LSTM (b) NVDA Stock-GNN-LSTM (c) SIRI Stock-GNN-LSTM (d) ZBH stock -GNN-LSTM (e) ABT stock -GNN-LSTM (f) ABT stock-GNN-Naive (g) NVDA stock-GNN-Naive (h) NFLX stock-GNN-Naive.

volatility. For example, clustering using fundamental ratios, which include earnings-to-price (EPS) ratios or other financial indicators, can give more detailed information on stock clusters.

V. CONCLUSION

Due to the spontaneous development of the social economy and the adaptive nature of stock market prices, the decision-making regarding investment in stocks is quite challenging. Investor-employed decision support systems includ-

ing stock trading, stock forecasting, and stock movement prediction are employed to make more informed trading decisions. However, most of the studies view the stocks as single and independent entities and then design a forecasting-based model. Such methodologies ignore the stock-to-stock relationships. Opposing that, some recent studies have attempted to model it using state-of-the-art Graph Neural Networks, however, graphs depending upon some solid criteria is one of the major research questions. Although, some heuristic methods have been designed but they have limited scope.

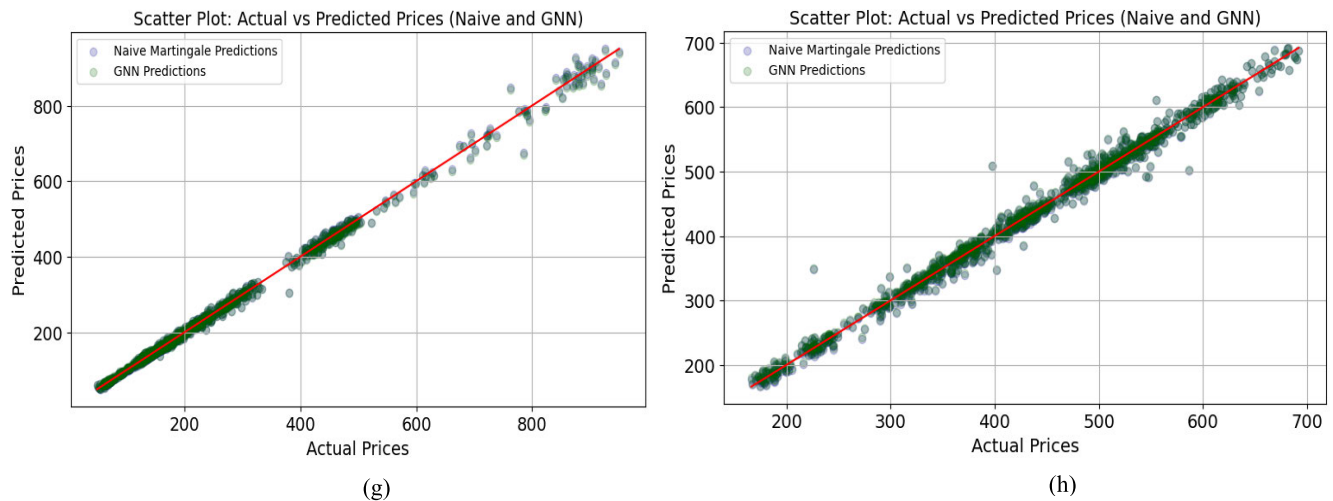


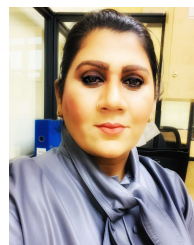
FIGURE 18. (Continued.) Scatter-plots of actual and predicted stock prices (a) NFLX Stock-GNN-LSTM (b) NVDA Stock-GNN-LSTM (c) SIRI Stock-GNN-LSTM (d) ZBH stock -GNN-LSTM (e) ABT stock -GNN-LSTM (f) ABT stock-GNN-Naive (g) NVDA stock-GNN-Naive (h) NFLX stock-GNN-Naive.

This study attempted to address such challenges and gaps and design a new variant of graph neural network referred to as multi-cluster graph (MCG), in which graphs are constructed using novel three clustering methodologies. The experimental results and findings indicate the proposed MCG model has very low error values and is scalable to large indices such as the S&P500 index containing 500 companies. In the future, dynamic stock relations approaches will be designed in combination with reinforcement learning to make the MCG model more adaptable to the evolving conditions of stock market prices.

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